

Engineers were forced to prioritise repairs and justify budgets without reliable data on crack depth, material loss, or surface deterioration rates. Imported inspection systems proved either too rigid or too costly for Indian conditions, exposing a clear gap between infrastructure investment and the quality of actionable data. They addressed this gap by developing their own platform, combining precision hardware, indigenous electronics, and domain expertise to transform infrastructure inspection into a measurable, data-driven engineering process.

Unlike conventional inspection approaches based on manual surveys or camera-only AI, they deliver survey-grade, three-dimensional measurements of surface defects while operating at highway speeds. This enables infrastructure owners to make informed maintenance and repair decisions. Simply put, their systems allow engineers to see infrastructure the way machines do—accurately, consistently, and in true 3D.

The system measures the length, width, depth, and area of cracks, potholes, rutting, patches, and surface deformation at sub-millimetre resolution. Even a 1mm difference can change fault classification. Traditional inspections and 2D AI lack depth, while most lidar systems offer only centimetre-level resolution.

The platform integrates 3D laser triangulation, high-speed specialised cameras, precision optics, embedded electronics, sensor fusion, and AI-based analytics into a single inspection system. It captures and processes survey-grade data at highway speeds of up to 100km/hr, covering up to 400km per day with accuracy of around 0.1mm. Inspection cycles that would take weeks using manual methods can be completed within days.

Maintaining sub-millimetre accuracy at highway speeds requires overcoming vibration, motion, heat,



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dust, and rain. So they designed the entire sensing, synchronisation, and processing pipeline in-house, integrating laser profilers, RGB cameras, GNSS, IMUs, and encoders across 17-18 channels, with custom electronics and hardware clocking to ensure precise time alignment, laser-camera calibration, and motion compensation.

From an electronics perspective, the platform functions as a high-speed data acquisition and processing system built for harsh environments. They design their own power-management boards, signal-conditioning PCBs, and data interfaces to handle gigabytes of data per second from laser profilers and cameras. Multiple communication standards, including USB, RS-232, and proprietary interfaces, operate simultaneously, while ruggedised enclosures and thermal design ensure stable operation in dust, rain, and extreme heat. A built-in 'Data Guard' feature continuously monitors sensor health and alerts operators in real time if any data stream fails, preventing costly data loss.

All core design and development is handled in-house by a multidisciplinary electronics, optics, and AI team, with external input only for

specialised mechanical and regulatory areas.

Mechanical fabrication is outsourced, while assembly, electronics integration, optics handling, and calibration are completed in-house at their Mumbai facility, enabling tight quality control and faster iteration.

Commercially, the startup is establishing a presence in the infrastructure inspection market. They have sold four systems to date, with three deployed and one under delivery. They reported a turnover of approximately ₹68 million in the last financial year and project revenues of around ₹130 million this year. With a team of about 30 employees, they operate on a direct B2B sales model, engaging customers through demonstrations, industry events, and technical discussions. Interest from overseas markets is also emerging.

The startup is now expanding their ecosystem through partnerships. In parallel, they plan to collaborate with academic institutions on joint research and development to validate and advance technical challenges related to infrastructure management. These collaborations will explore deeper use of laser profilers, camera-based systems, and advanced sensing technologies for applications such as road safety enhancement, urban planning, and accident reduction. Partnership expansion and academic collaboration form part of their near-term roadmap, with several initiatives expected over the coming year.

Looking ahead, their roadmap includes new inspection systems for railways, advanced mapping solutions, and specialised inspection tools for assets such as cables and wire ropes in 2026. They also plan to expand AI capabilities from defect detection to predictive analytics that forecast deterioration, prioritise maintenance, and simulate long-term outcomes. **EFY**

—Ananksha Sondhi Gaur